

Foundation (Jalapeno)

- Q1.** Simplify $2^3 \cdot 2^4$
(A) 2^7
(B) 2^6
(C) 2^5
(D) 2^3
(E) 2^2
- Q2.** What is the value of x in $x^2 = 16$?
(A) 2
(B) 4
(C) -2
(D) -4
(E) 8
- Q3.** A spaceship travels at a speed of 3^2 km/s. How far will it travel in 2 seconds?
(A) 9 km
(B) 18 km
(C) 12 km
(D) 6 km
(E) 4 km
- Q4.** Simplify $(2^2)^3$
(A) 2^5
(B) 2^6
(C) 2^8
(D) 2^4
(E) 2^9
- Q5.** What is the value of y in $y^3 = 64$?
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6
- Q6.** A lab experiment requires 2^4 liters of oxygen. How many liters are needed for 3 experiments?
(A) 48 liters
(B) 24 liters
(C) 16 liters
(D) 12 liters
(E) 8 liters
- Q7.** Simplify $2^6 2^2$
(A) 2^3
(B) 2^4
(C) 2^5
(D) 2^8
(E) 2^9
- Q8.** The distance between two planets is 3^2 light years. How many light years will a spaceship travel if it goes from one planet to the other and back?
(A) 12 light years
(B) 18 light years
(C) 9 light years
(D) 6 light years
(E) 15 light years

Open Ended Questions (FRQ)

- Q9.** A rocket is launched into space with an initial velocity of 2^3 m/s. It accelerates at a rate of 2^2 m/s² for 3 seconds. Calculate the final velocity of the rocket. Show all work and simplify your answer.
- Q10.** The volume of a planet is given by the equation $V = \frac{4}{3}\pi r^3$, where r is the radius of the planet. If the radius of a planet is 2^2 km, calculate the volume of the planet in cubic kilometers. Show all work and simplify your answer.

Chef's Tips

- Emphasize the importance of applying exponent rules correctly
- Use real-world examples to illustrate the relevance of exponents in space and science
- Encourage students to check their work and simplify their answers